A=10 B=5 C=7

① Five students (P_0 to P_4)

A, B.

Processes	Allocation A B C	Max A B C	Avaliable A B C	Need A B C
P_0	0 1 0	7 5 3	3 3 2	7 4 3
P_1	2 0 0	3 2 2	5 3 2	1 2 2
P_2	3 0 2	9 0 2	7 4 3	6 0 0
P_3	2 1 1	2 2 2	7 4 5	0 1 1
P_4	0 0 2	4 3 3	7 5 5	4 3 1
			10 5 7	

~~Avaliable~~ = Total − Allocation

A = 10 − 7 = 3

B = 5 − 2 = 3

C = 7 − 5 = 2

Need = Max − Allocation

Need ≤ Avaliable

$P_1 \rightarrow P_3 \rightarrow P_4 \rightarrow P_0 - P_2$

② Avaliable Resources.

A=10, B=7, C=10

Process	Allocation A B C	Max A B C	Avaliable A B C	Need A B C
P_0	1 2 2	3 3 2	4 2 3	2 1 0
P_1	2 0 0	2 2 2	5 4 5	0 2 2
P_2	2 1 1	4 3 3	7 4 5	2 2 2
P_3	0 2 1	1 3 5	9 5 6	1 1 4
P_4	1 0 3	3 3 3	9 7 7	2 3 0
			10 7 10	

Avaliable = Total - Allocation

A = 10 - 6 = 4

B = 7 - 5 = 2

C = 10 - 7 = 3

$P_0 \rightarrow P_1 \rightarrow P_2 \rightarrow P_3 \rightarrow P_4$

(3) Avaliable A = 10, B = 6, C = 9.

	Allocation			Max			Avaliable			Need		
	A	B	C	A	B	C	A	B	C	A	B	C
P_0	0	1	0	6	5	3	3	4	3	6	4	3
P_1	2	0	0	3	2	2	5	4	3	1	2	2
P_2	3	0	3	6	0	3	8	4	6	3	0	0
P_3	2	1	1	2	2	2	10	5	7	0	1	1
P_4	0	0	2	4	3	3	10	5	9	4	3	1
							10	6	9			

Need = Max - Allocation Need ≤ Avaliable.

A = 10 - 7 = 3

B = 6 - 2 = 4

C = 9 - 6 = 3

$P_1 \rightarrow P_2 \rightarrow P_3 \rightarrow P_4 \rightarrow P_0$

Avaliable A = 9, B = 5, C = 8

(4)

	Allocation			Max			Avaliable			Need		
	A	B	C	A	B	C	A	B	C	A	B	C
P_0	1	0	1	3	2	2	2	3	3	2	2	1
P_1	2	1	0	4	2	1	3	3	4	2	1	1
P_2	2	0	1	4	1	2	5	4	4	2	1	1
P_3	1	1	2	2	2	2	7	4	5	1	1	0
P_4	1	0	1	3	1	3	8	5	7	2	1	2
							9	5	8			

Avaliable

A = 9 - 7 = 2

B = 5 - 2 = 3

C = 8 - 5 = 3

$P_0 \rightarrow P_1 \rightarrow P_2 \rightarrow P_3 \rightarrow P_4$

⑤ Avaliable

A = 12, B = 7 C = 10

	Allocation			Max			Avaliable			Need		
	A	B	C	A	B	C	A	B	C	A	B	C
P_0	2	1	2	5	3	3	3	2	2	3	2	1
P_1	1	1	1	3	2	2	5	3	4	2	1	1
P_2	3	0	2	6	1	3	6	4	5	3	1	1
P_3	2	1	2	3	2	4	9	4	7	1	1	2
P_4	1	2	1	2	3	3	11	5	9	1	1	2
							12	7	10			

Avaliable = Total - Allocation

A = 12 - 9 = 3

B = 7 - 5 = 2

C = 10 - 8 = 2

$P_0 \rightarrow P_1 \rightarrow P_2 \rightarrow P_3 \rightarrow P_4$

$P_0 (3, 2, 1) \leq (3, 2, 2)$

$P_1 (2, 1, 1) \leq (5, 3, 4)$

$P_2 (3, 1, 1) \leq (6, 4, 8)$

$P_3 (1, 1, 2) \leq (9, 4, 7)$

$P_4 (1, 1, 2) \leq (11, 5, 9)$